

QUESTION NO#01:

```
#include <iostream>

using namespace std;

class Array2D {
    int rows, cols;
    int** data;

public:
    Array2D() {
        rows = cols = 0;
        data = nullptr;
    }

    Array2D(int r, int c) {
        rows = r;
        cols = c;
        data = new int*[rows];
        for (int i = 0; i < rows; i++)
            data[i] = new int[cols];
    }

    Array2D(const Array2D& obj) {
        rows = obj.rows;
```

```
    cols = obj.cols;
    data = new int*[rows];
    for (int i = 0; i < rows; i++) {
        data[i] = new int[cols];
        for (int j = 0; j < cols; j++)
            data[i][j] = obj.data[i][j];
    }
}
```

```
Array2D& operator=(const Array2D& obj) {
    if (this != &obj) {
        for (int i = 0; i < rows; i++)
            delete[] data[i];
        delete[] data;

        rows = obj.rows;
        cols = obj.cols;
        data = new int*[rows];
        for (int i = 0; i < rows; i++) {
            data[i] = new int[cols];
            for (int j = 0; j < cols; j++)
                data[i][j] = obj.data[i][j];
        }
    }
    return *this;
}
```

```
Array2D operator+(const Array2D& obj) {  
    Array2D res(rows, cols);  
    for (int i = 0; i < rows; i++)  
        for (int j = 0; j < cols; j++)  
            res.data[i][j] = data[i][j] + obj.data[i][j];  
    return res;  
}
```

```
Array2D operator-(const Array2D& obj) {  
    Array2D res(rows, cols);  
    for (int i = 0; i < rows; i++)  
        for (int j = 0; j < cols; j++)  
            res.data[i][j] = data[i][j] - obj.data[i][j];  
    return res;  
}
```

```
Array2D operator*(const Array2D& obj) {  
    Array2D res(rows, obj.cols);  
    for (int i = 0; i < rows; i++)  
        for (int j = 0; j < obj.cols; j++) {  
            res.data[i][j] = 0;  
            for (int k = 0; k < cols; k++)  
                res.data[i][j] += data[i][k] * obj.data[k][j];  
        }  
    return res;  
}
```

```
}
```

```
int* operator[](int i) {  
    return data[i];  
}
```

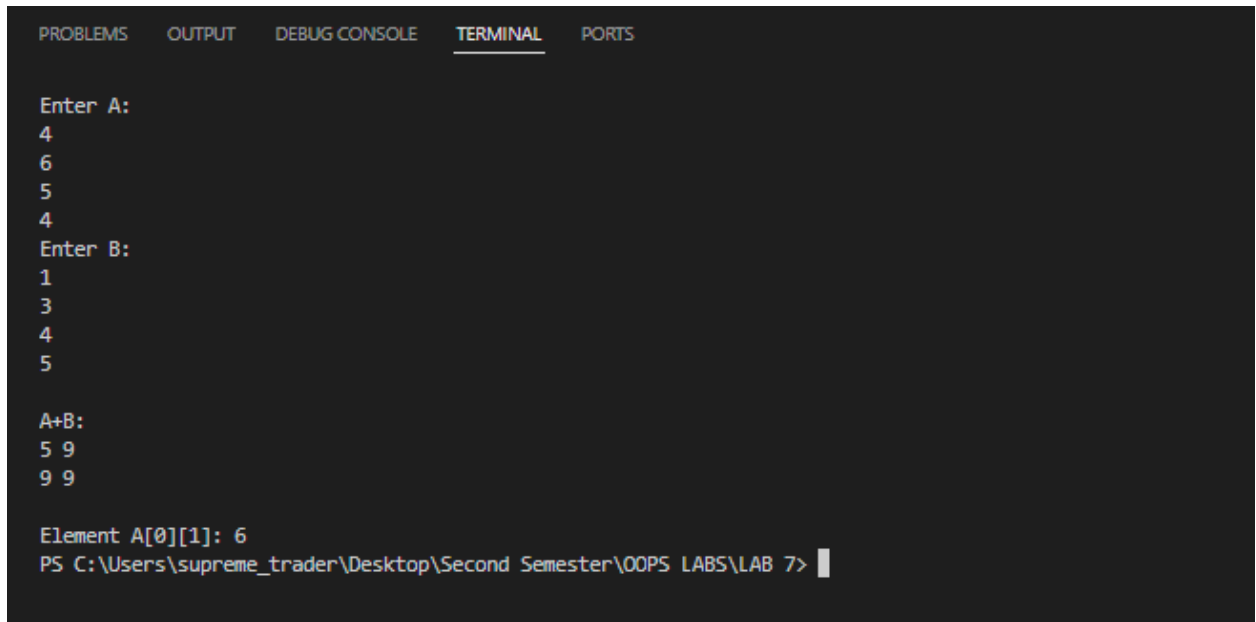
```
void input() {  
    for (int i = 0; i < rows; i++)  
        for (int j = 0; j < cols; j++)  
            cin >> data[i][j];  
}
```

```
void display() {  
    for (int i = 0; i < rows; i++) {  
        for (int j = 0; j < cols; j++)  
            cout << data[i][j] << " ";  
        cout << endl;  
    }  
}
```

```
~Array2D() {  
    for (int i = 0; i < rows; i++)  
        delete[] data[i];  
    delete[] data;  
}  
};
```

```
int main() {  
    Array2D A(2,2), B(2,2);  
  
    cout << "Enter A:\n";  
    A.input();  
  
    cout << "Enter B:\n";  
    B.input();  
  
    Array2D C = A + B;  
    cout << "\nA+B:\n";  
    C.display();  
  
    cout << "\nElement A[0][1]: " << A[0][1] << endl;  
  
    return 0;  
}
```

OUTPUT:



The screenshot shows a C++ IDE with a terminal window. The terminal has tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL (selected), and PORTS. The output text is as follows:

```
Enter A:
4
6
5
4
Enter B:
1
3
4
5

A+B:
5 9
9 9

Element A[0][1]: 6
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 7> |
```

QUESTION NO#02:

```
#include <iostream>
```

```
using namespace std;
```

```
class Array2D {
```

```
    int rows, cols;
```

```
    int** data;
```

```
public:
```

```
    Array2D(int r, int c) {
```

```
        rows = r;
```

```
        cols = c;
```

```
data = new int*[rows];  
for (int i = 0; i < rows; i++)  
    data[i] = new int[cols];  
}
```

```
void input() {  
    for (int i = 0; i < rows; i++)  
        for (int j = 0; j < cols; j++)  
            cin >> data[i][j];  
}
```

```
bool search(int target) {  
    int left = 0, right = rows * cols - 1;  
  
    while (left <= right) {  
        int mid = (left + right) / 2;  
  
        int r = mid / cols;  
        int c = mid % cols;  
  
        if (data[r][c] == target)  
            return true;  
        else if (data[r][c] < target)  
            left = mid + 1;  
        else  
            right = mid - 1;  
    }  
}
```

```
    }  
    return false;  
}  
  
~Array2D() {  
    for (int i = 0; i < rows; i++)  
        delete[] data[i];  
    delete[] data;  
}  
};  
  
int main() {  
    Array2D A(3,4);  
  
    cout << "Enter matrix:\n";  
    A.input();  
  
    int target;  
    cout << "Enter target: ";  
    cin >> target;  
  
    if (A.search(target))  
        cout << "True";  
    else  
        cout << "False";
```

```
    return 0;  
}
```

OUTPUT:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
  
Enter matrix:  
1  
3  
5  
7  
10  
11  
16  
20  
23  
30  
34  
60  
Enter target: 3  
True  
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 7>
```



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  Quick-Run + v  
  
Enter matrix:  
1  
3  
5  
7  
10  
11  
16  
20  
23  
30  
34  
60  
Enter target: 13  
False  
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 7>
```

QUESTION NO#03:

```
#include <iostream>

using namespace std;

class Employee {
private:
    string name;
    int id;
    string designation;
    double salary;

public:
    Employee(string n, int i, string d, double s) {
        name = n;
        id = i;
        designation = d;
        salary = s;
    }

    void display() {
        cout << "Name: " << name << endl;
        cout << "ID: " << id << endl;
        cout << "Designation: " << designation << endl;
        cout << "Salary: " << salary << endl;
    }
};
```

```
}

friend class Payroll;

};

class Payroll {
public:
    void updateSalary(Employee &e, double increment) {
        e.salary += increment;
    }
};

int main() {
    Employee emp("Ali", 101, "Manager", 50000);

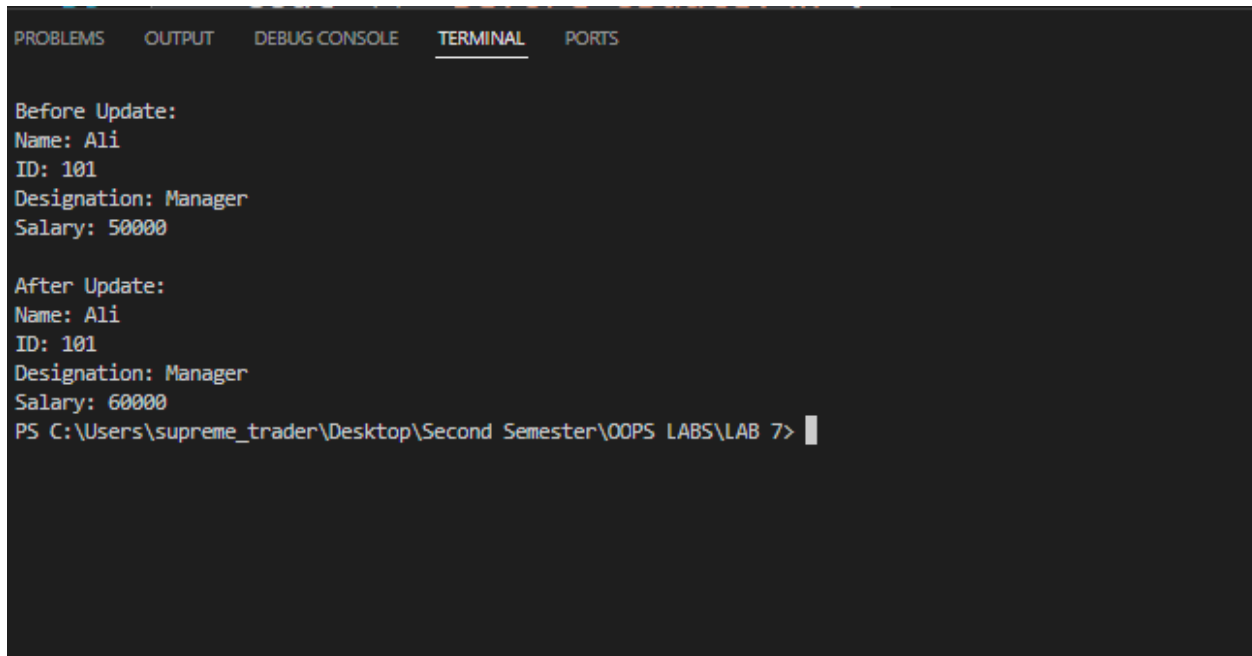
    cout << "Before Update:\n";
    emp.display();

    Payroll p;
    p.updateSalary(emp, 10000);

    cout << "\nAfter Update:\n";
    emp.display();

    return 0;
}
```

OUTPUT:

A screenshot of a C++ IDE's terminal window. The terminal has tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL (which is active), and PORTS. The output text is as follows:

```
Before Update:
Name: Ali
ID: 101
Designation: Manager
Salary: 50000

After Update:
Name: Ali
ID: 101
Designation: Manager
Salary: 60000
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 7> |
```

QUESTION NO#04:

```
#include <iostream>

using namespace std;

class Employee {
private:
    string name;
    int id;
    string designation;
    double salary;
```

public:

```
Employee(string n, int i, string d, double s) {  
    name = n;  
    id = i;  
    designation = d;  
    salary = s;  
}
```

```
void display() {  
    cout << "Name: " << name << endl;  
    cout << "ID: " << id << endl;  
    cout << "Designation: " << designation << endl;  
    cout << "Salary: " << salary << endl;  
}
```

```
friend void updateSalary(Employee &e, double increment);  
};
```

```
void updateSalary(Employee &e, double increment) {  
    e.salary += increment;  
}
```

```
int main() {  
    Employee emp("Ali", 101, "Manager", 50000);  
  
    cout << "Before Update:\n";
```

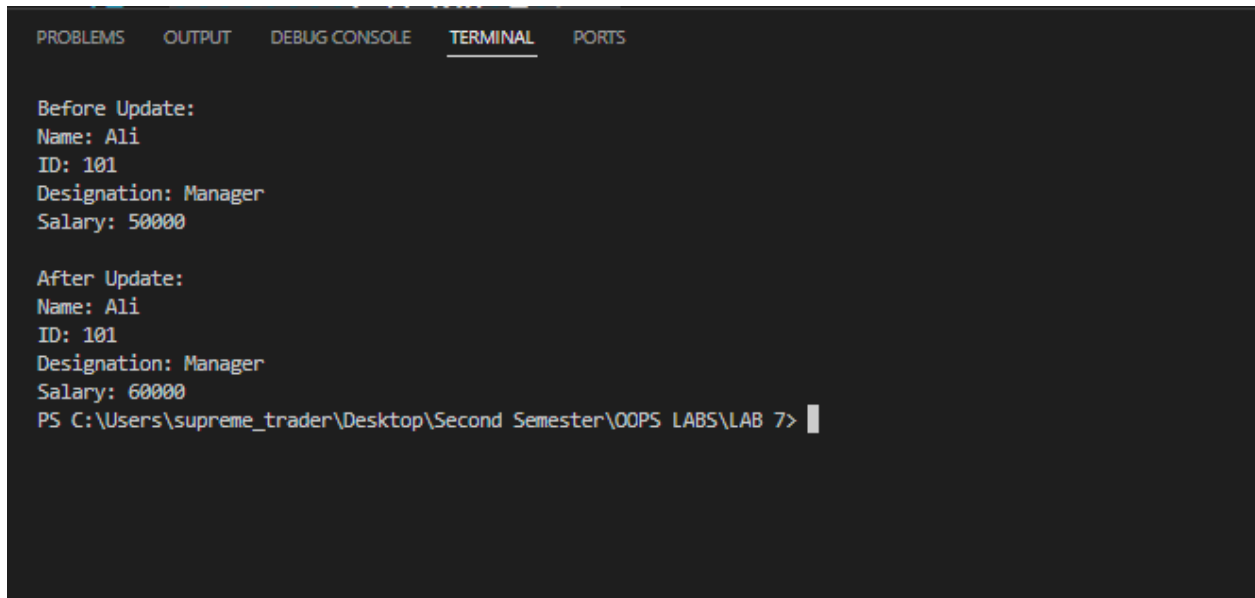
```
emp.display();

updateSalary(emp, 10000);

cout << "\nAfter Update:\n";
emp.display();

return 0;
}
```

OUTPUT:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Before Update:
Name: Ali
ID: 101
Designation: Manager
Salary: 50000

After Update:
Name: Ali
ID: 101
Designation: Manager
Salary: 60000
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 7> |
```